

Slope of Linear Functions from Graphs

Jonathan R. Bacolod

Slope

The **slope** of a line measures its **rate of change**.

Slope as Rate of Change

Slope tells how much the y -value changes for every unit change in the x -value.

Slope Formula

The slope of a line is

$$m = \frac{\text{rise}}{\text{run}} = \frac{\text{vertical change}}{\text{horizontal change}}$$

Positive Slope

A **positive slope** means the graph rises from left to right.

Negative Slope

A **negative slope** means the graph falls from left to right.

Zero Slope

A **zero slope** corresponds to a horizontal line.

Undefined Slope

An **undefined slope** corresponds to a vertical line.

Examples

Example 1

A line passes through $(0, 3)$ and $(2, 0)$.

$$m = \frac{0 - 3}{2 - 0} = -\frac{3}{2}$$

The slope is

$$\boxed{-\frac{3}{2}}$$

Example 2

A line passes through $(0, 1)$ and $(4, 5)$.

$$m = \frac{5 - 1}{4 - 0} = 1$$

The slope is

$$\boxed{1}$$

Example 3

A horizontal line passes through $(1, -1)$ and $(5, -1)$.

$$m = \frac{-1 - (-1)}{5 - 1} = 0$$

The slope is

$$\boxed{0}$$

Example 4

A vertical line passes through $(2, 1)$ and $(2, 5)$.

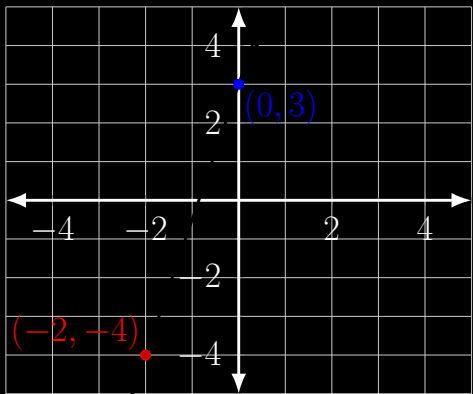
$$m = \frac{5 - 1}{2 - 2} = \frac{4}{0}$$

The slope is **undefined**.

Practice Exercises

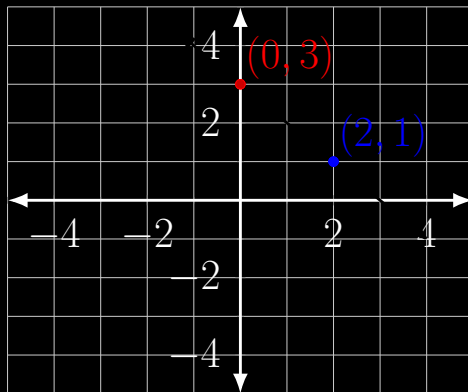
Practice 1

For the graph, determine the **slope** of the function.



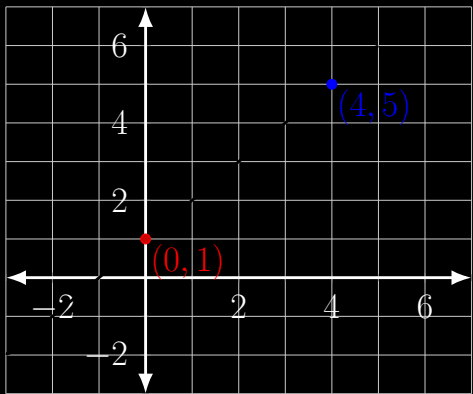
Practice 2

For the graph, determine the **slope** of the function.



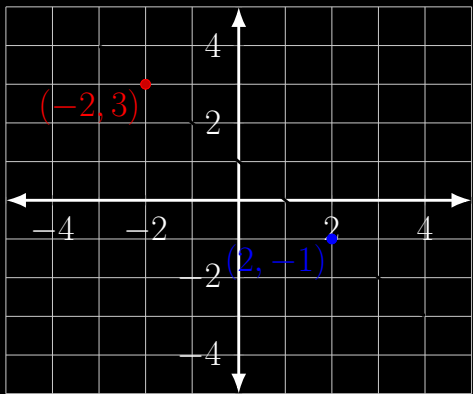
Practice 3

For the graph, determine the **slope** of the function.



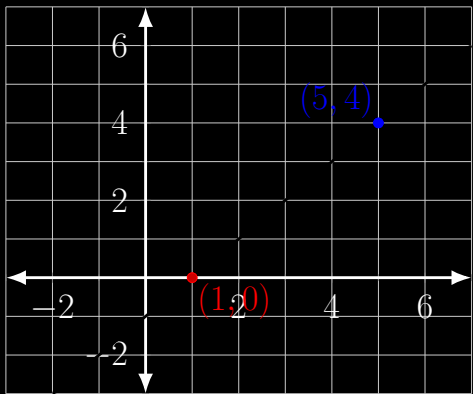
Practice 4

For the graph, determine the **slope** of the function.



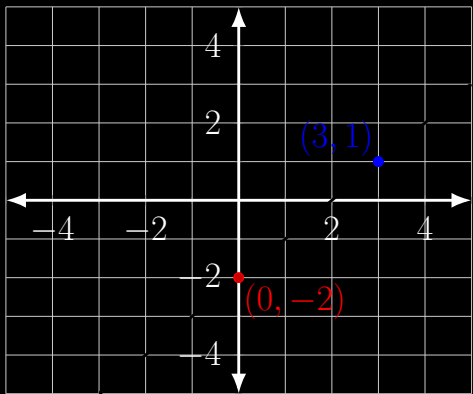
Practice 5

For the graph, determine the **slope** of the function.



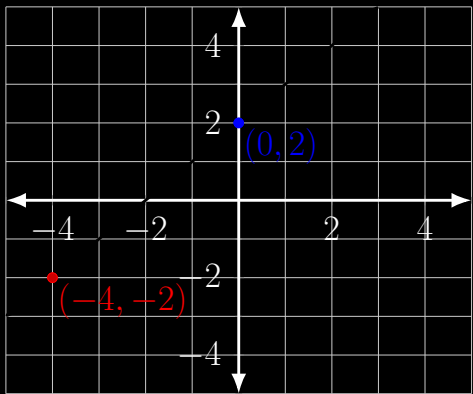
Practice 6

For the graph, determine the **slope** of the function.



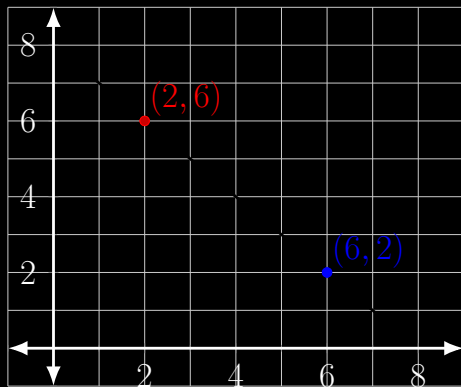
Practice 7

For the graph, determine the **slope** of the function.



Practice 8

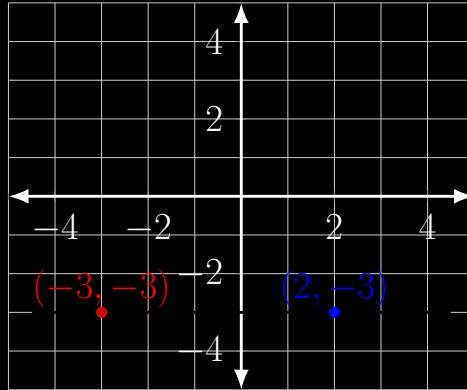
For the graph, determine the **slope** of the function.



Activities

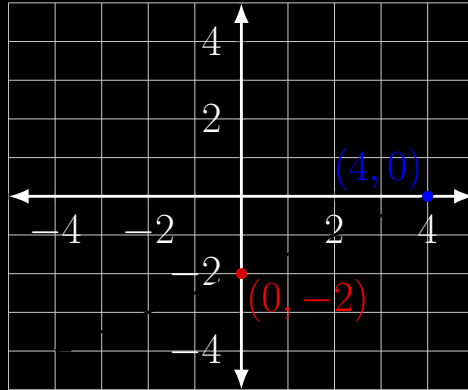
Activity 1

For the graph, determine the **slope** of the function.



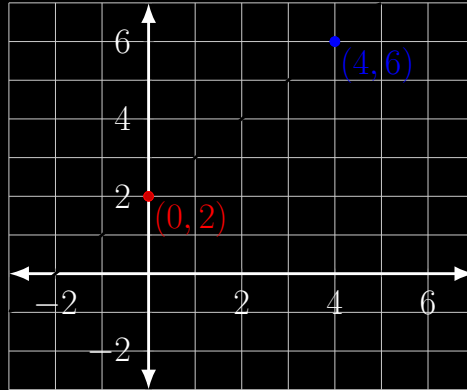
Activity 2

For the graph, determine the **slope** of the function.



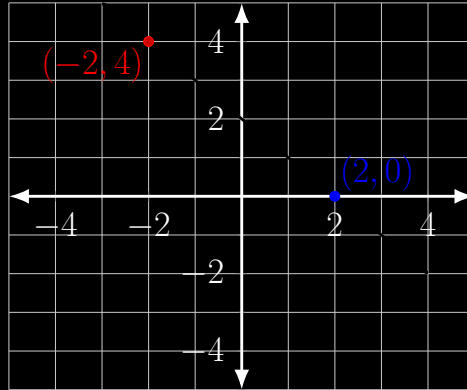
Activity 3

For the graph, determine the **slope** of the function.



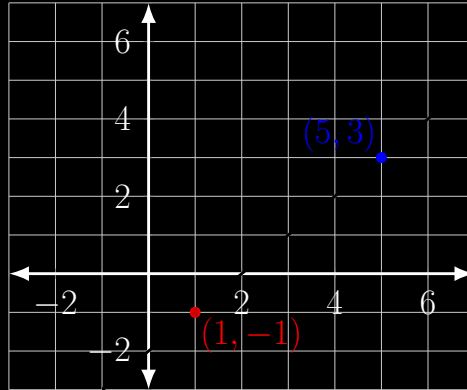
Activity 4

For the graph, determine the **slope** of the function.



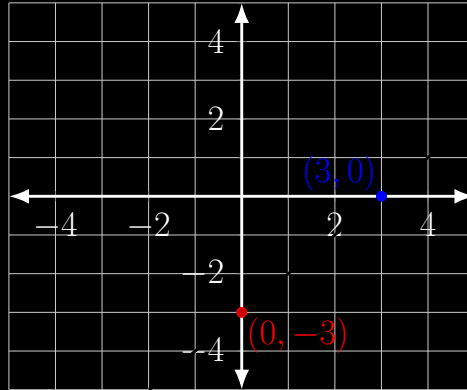
Activity 5

For the graph, determine the **slope** of the function.



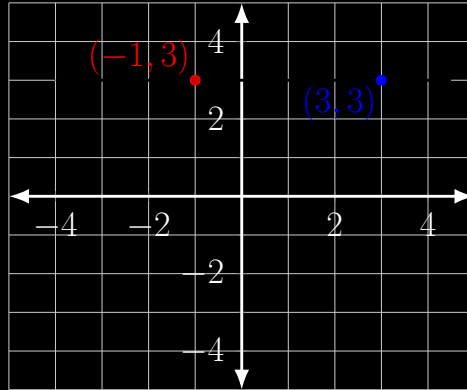
Activity 6

For the graph, determine the **slope** of the function.



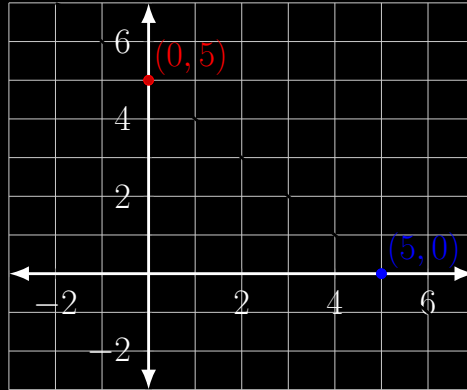
Activity 7

For the graph, determine the **slope** of the function.



Activity 8

For the graph, determine the **slope** of the function.



Thank you!